STA2023 Exam 2 — SIDE A — THE PRACTICE EXAM, SOLVED (the real exam models it)

■ GREEN = the question ■ RED = rule / answer ■ BLACK = the work ■ BLUE = if they switch it up

Every part: formula → plug in → arithmetic → box the answer. Probabilities to 4 decimals. No work = no credit.

Q1 (6 pts) — Two-way table: or / and / given

"The table below summarizes the employment status of 78 residents. Answer each question. Show your work."

Status	Males	Females	Row total
Full-time	22	18	40
Part-time	6	10	16
Unemployed	3	2	5
Not in labor force	5	12	17
Column total	**36**	**42**	**78**

First: add the row and column totals in the margins. Every answer is then **a count ÷ the group you are picking from**.

(a) probability that a randomly selected resident is male? one category → its total ÷ grand total

$$P(\text{male}) = \frac{36}{78} = 0.4615$$

(b) …is part-time? a row → add both cells, then ÷ 78

$$P(\text{part-time}) = \frac{6+10}{78} = \frac{16}{78} = 0.2051$$

(c) …is male OR unemployed?

OR: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ **— adding counts the overlap (male AND unemployed = 3) twice, so take it out once**

$$P(\text{M or U}) = \frac{36}{78} + \frac{5}{78} - \frac{3}{78} = \frac{38}{78} = 0.4872$$

Watch out: forgetting the overlap gives 41/78 = 0.5256 × (check: 36 males + 2 unemployed women = 38 ✓)

(d) …is female AND not in the labor force? AND in a table = the ONE cell where the row and column meet, ÷ 78

$$P(\text{F and NLF}) = \frac{12}{78} = 0.1538$$

(e) …is part-time GIVEN that the resident is female?

GIVEN: $P(B \mid A) = \dfrac{N(A \text{ and } B)}{N(A)}$ **— A = the given event (the word after "given"): divide by that group's total, 42, not 78**

$$P(\text{PT} \mid \text{Female}) = \frac{10}{42} = 0.2381$$

Watch out: 10/78 = 0.1282 is the classic error.

If they switch it up: NOT male → $1 - \frac{36}{78} = \frac{42}{78} = 0.5385$ · **female GIVEN part-time** → $\frac{10}{16} = 0.6250$ (direction matters) · **independent?** P(PT | F) = 0.2381 ≠ P(PT) = 0.2051 → dependent

Q2 (2 pts) — Valid discrete probability distribution?

"Determine whether the following distribution represents a valid discrete probability distribution. If not, explain why." x: 0, 1, 2, 3, 4 with P(x): 0.34, 0.21, 0.13, 0.04, 0.01

Valid only if (1) every $0 \le P(x) \le 1$ AND (2) $\sum P(x) = 1$. Write BOTH checks.

Rule 1: 0.34, 0.21, 0.13, 0.04, 0.01 are all between 0 and 1 ✓

Rule 2: add them:

$$0.34 + 0.21 + 0.13 + 0.04 + 0.01 = 0.73 \neq 1 \ \times$$

NOT a valid probability distribution — the probabilities add to 0.73, not 1.

If they switch it up: a negative P(x) or one above 1 → not valid even if the sum is 1 · sum above 1 → not valid · **"find the missing probability"** → 1 − (sum of the others): if P(4) were blank, P(4) = 1 − 0.72 = 0.28

Q3 (8 pts) — Stanley Cup: discrete random variable

"The following data represent the number of games played, X, in the Stanley Cup before a champion was determined." x: 4, 5, 6, 7 with frequency 20, 20, 26, 17. (a) Construct a probability distribution for X. (b) Graph it. (c) Compute and interpret the mean. (d) Compute the standard deviation.

(a) construct the probability distribution

$P(x) = \dfrac{\text{frequency}}{\text{total}}$ **total = 20 + 20 + 26 + 17 = 83**

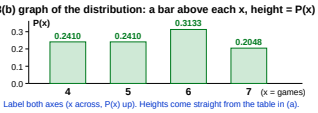
$$P(4) = \frac{20}{83} = 0.2410 \qquad P(5) = \frac{20}{83} = 0.2410$$

$$P(6) = \frac{26}{83} = 0.3133 \qquad P(7) = \frac{17}{83} = 0.2048$$

Check: 0.2410 + 0.2410 + 0.3133 + 0.2048 = 1.0001 ≈ 1 ✓ (rounding).

(b) graph the discrete probability distribution

x across, P(x) up; a vertical bar (or line) above each x whose height is its probability; label both axes.



(c) compute and interpret the mean of X

$\mu_X = \sum x \cdot P(x)$ **— a weighted average: each x counts as much as its probability**

$$\mu = 4(0.2410) + 5(0.2410) + 6(0.3133) + 7(0.2048)$$
$$= 0.9640 + 1.2050 + 1.8798 + 1.4336 = 5.48 \text{ games}$$

μ ≈ 5.48 games **Interpretation (part of the points):** over many Stanley Cup series, the average length is about 5.48 games.

(d) compute the standard deviation of X

$\sigma_X = \sqrt{\sum (x-\mu)^2 \cdot P(x)}$ **with μ = 5.48: subtract, square, × P(x), add, √**

x	x − μ	(x − μ)²	(x − μ)² · P(x)
4	−1.48	2.1904	2.1904(0.2410) = 0.5279
5	−0.48	0.2304	0.2304(0.2410) = 0.0555
6	0.52	0.2704	0.2704(0.3133) = 0.0847
7	1.52	2.3104	2.3104(0.2048) = 0.4732

$$\sigma^2 = 0.5279 + 0.0555 + 0.0847 + 0.4732 = 1.1413$$
$$\sigma = \sqrt{1.1413} = 1.07 \text{ games}$$

σ ≈ 1.07 games

Watch out: don't stop at the variance 1.1413; no ÷ N here — the P(x) do the averaging.

If they switch it up: P(at least 6 games) = P(6) + P(7) = 0.3133 + 0.2048 = 0.5181 · **P(fewer than 6)** = P(4) + P(5) = 0.4820 · **variance** = σ² = 1.1413 (stop before the √) · **expected value E(X)** = the mean

Q4 (8 pts) — Driving age: binomial

"According to a Gallup poll, 60% of U.S. women 18 years old or older stated that the minimum driving age should be 18. In a random sample of 10 U.S. women, find the probability that (a) exactly 3, (b) fewer than 3, (c) between 1 and 3 (inclusive) believe this. (d) For a random sample of 200, find the expected number and standard deviation."

Binomial: fixed n = 10 yes/no trials, independent, same p. **n = 10, p = 0.60, 1 − p = 0.40, X = number who say yes.**

$$P(x) = {}_nC_x\, p^x (1-p)^{n-x} \qquad {}_nC_x = \frac{n!}{x!\,(n-x)!}$$

(a) exactly 3 believe the minimum driving age should be 18

$${}_{10}C_3 = \frac{10!}{3!\,7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$
$$P(3) = {}_{10}C_3\,(0.60)^3(0.40)^7 = 120(0.216)(0.0016384) = 0.0425$$

0.0425

Watch out: the exponent on 0.40 is n − x = 7, not 3; keep all decimals until the end.

(b) fewer than 3 believe… "fewer than 3" = 0, 1, 2 — the 3 is OUT. Add the three separate probabilities.

$$P(0) = {}_{10}C_0(0.6)^0(0.4)^{10} = 1(1)(0.0001049) = 0.0001$$
$$P(1) = {}_{10}C_1(0.6)^1(0.4)^9 = 10(0.6)(0.0002621) = 0.0016$$
$$P(2) = {}_{10}C_2(0.6)^2(0.4)^8 = 45(0.36)(0.0006554) = 0.0106$$
$$P(X<3) = 0.0001 + 0.0016 + 0.0106 = 0.0123$$

0.0123

(c) between 1 and 3 (inclusive) believe… inclusive → x = 1, 2, 3. Reuse (a) and (b) — write "from part (a)".

$$P(1 \le X \le 3) = P(1) + P(2) + P(3) = 0.0016 + 0.0106 + 0.0425 = 0.0547$$

0.0547

(d) sample of 200: expected number and standard deviation

binomial shortcuts: $\mu = np$, $\sigma = \sqrt{np(1-p)}$

$$\mu = np = 200(0.60) = 120$$
$$\sigma = \sqrt{np(1-p)} = \sqrt{200(0.60)(0.40)} = \sqrt{48} = 6.93$$

μ = 120, σ ≈ 6.93 (48 is the variance — take the √)

If they switch it up: at least 3 = 1 − P(X ≤ 2) = 1 − 0.0123 = 0.9877 · **at most 3** = P(X ≤ 2) + P(3) = 0.0123 + 0.0425 = 0.0548 · **more than 3** = 1 − 0.0548 = 0.9452 · **"would 130 of 200 be unusual?"** np(1 − p) = 48 ≥ 10 ✓ → usual range μ ± 2σ = 120 ± 2(6.93) = 106.1 to 133.9 → 130 is inside → not unusual

Q5 (5 pts) — Daily temperature: uniform

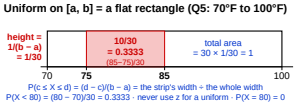
"The daytime temperature X (°F) in a coastal city during summer is uniformly distributed between 70°F and 100°F. (a) Sketch or describe the graph of the uniform pdf. (b) Find P(75 ≤ X ≤ 85). (c) Find P(X < 80)."

a = 70, b = 100, width b − a = 30. $\text{height} = \dfrac{1}{b-a}$ $P(c \le X \le d) = \dfrac{d-c}{b-a}$ **= the piece's share of the whole width. No z-scores here.**

(a) sketch / describe the pdf

$$\text{height} = \frac{1}{b-a} = \frac{1}{100-70} = \frac{1}{30} \qquad \text{area} = 30 \cdot \frac{1}{30} = 1 \ \checkmark$$

a flat line (rectangle) at height 1/30 from 70 to 100, and 0 everywhere else



(b) P(75 ≤ X ≤ 85)

$$P(75 \le X \le 85) = \frac{85-75}{100-70} = \frac{10}{30} = 0.3333$$

0.3333

(c) P(X < 80)

$$P(X < 80) = \frac{80-70}{100-70} = \frac{10}{30} = 0.3333$$

0.3333

Watch out: P(X ≤ 80) is the same number — one point has no width, P(X = 80) = 0.

If they switch it up: P(X > 90) = $\frac{100-90}{30} = 0.3333$ · **"20% of days are cooler than what temperature?"** → $x = a + P(b-a) = 70 + 0.20(30) = 76°\text{F}$ · **P(X = 80)** = 0

Q6 (6 pts) — Standard normal: the table gives the area to the LEFT

"Use the standard normal table. (a) Find the area to the left of z = −1.10. (b) Find the area between z = −0.60 and z = 1.45. (c) X is normally distributed with μ = 80 and σ = 10. Find P(70 < X < 95)."

LEFT = table value · RIGHT = 1 − table · BETWEEN = (left of the bigger z) − (left of the smaller z) · $z = \dfrac{x-\mu}{\sigma}$ **rounded to 2 decimals**

(a) area to the left of z = −1.10

Row −1.1, column .00 → read the entry: **0.1357**

Watch out: positive-only table? symmetry: 1 − (left of +1.10) = 1 − 0.8643 = 0.1357.

(b) area between z = −0.60 and z = 1.45

left of 1.45 (row 1.4, col .05) = 0.9265; left of −0.60 = 0.2743.

$$\text{between} = 0.9265 - 0.2743 = 0.6522$$

0.6522

Watch out: subtract the AREAS, never the z's; a negative answer means you subtracted the wrong way.

(c) μ = 80, σ = 10: P(70 < X < 95)

$$z_1 = \frac{70-80}{10} = -1.00 \ \rightarrow \ \text{left} = 0.1587$$
$$z_2 = \frac{95-80}{10} = 1.50 \ \rightarrow \ \text{left} = 0.9332$$
$$P(70 < X < 95) = 0.9332 - 0.1587 = 0.7745$$

0.7745

If they switch it up: right of z = 1.45 = 1 − 0.9265 = 0.0735 · **P(X > 95)** = 1 − 0.9332 = 0.0668 · **z with area 0.10 to its LEFT** → closest entry 0.1003 → z = −1.28 · **z with area 0.025 to its RIGHT** → left area 0.975 → z = 1.96

Q7 (8 pts) — Wechsler IQ: normal with μ = 100, σ = 15

"The Wechsler Intelligence Scale for Children is approximately normally distributed with mean μ = 100 and standard deviation σ = 15. (a) P(score above 125)? (b) P(score below 90)? (c) What proportion score between 110 and 140? (d) What score places a child in the 98th percentile?"

Forward (a)–(c): $z = \dfrac{x-100}{15}$ **→ table → below: left · above: 1 − left · between: bigger left − smaller left. Backward (d): area → z (found INSIDE the table) →** $x = \mu + z\sigma$

(a) probability a test taker scores above 125

$$z = \frac{125-100}{15} = 1.67 \ \rightarrow \ \text{left} = 0.9525$$
$$P(X > 125) = 1 - 0.9525 = 0.0475$$

0.0475

Watch out: 0.9525 is the #1 wrong answer; 125 is above average → answer below 0.5.

(b) probability a test taker scores below 90

$$z = \frac{90-100}{15} = -0.67$$
$$P(X < 90) = \text{left of } -0.67 = 0.2514$$

0.2514 (keep the minus sign on z)

(c) proportion scoring between 110 and 140

$$z_1 = \frac{110-100}{15} = 0.67 \ \rightarrow \ \text{left} = 0.7486$$
$$z_2 = \frac{140-100}{15} = 2.67 \ \rightarrow \ \text{left} = 0.9962$$
$$P(110 < X < 140) = 0.9962 - 0.7486 = 0.2476$$

0.2476 (proportion = probability; 4 decimals)

(d) score at the 98th percentile

98th percentile = 98% score below it → area to the LEFT = 0.98. Find 0.98 INSIDE the table: 0.9798 (z = 2.05) is closer than 0.9803 (z = 2.06) → z = 2.05.

$$x = \mu + z\sigma = 100 + 2.05(15)$$
$$= 100 + 30.75 = 130.75 \approx 131$$

≈ 131 (130.75)

Watch out: never put 0.98 into the z formula — it is an area, not a score.

If they switch it up: top 5% → left 0.95 → z = 1.645 → x = 100 + 1.645(15) = 124.7 · **bottom 10%** → left 0.10 → z = −1.28 → x = 100 − 1.28(15) = 80.8 · **middle 95%** → z = ±1.96 → 70.6 to 129.4 · **is a score of 70 unusual?** z = −2.00 → 0.0228 < 0.05 → yes

STA2023 Exam 2 — SIDE B — RULES, FORMULAS, PICTURES

■ BLUE = what it means ■ RED = formula / rule ■ GREEN = example ■ BLACK = steps

SYMBOLS — what they mean

Symbol	Plain English
Σ	"add them all up"
N / n	how many values (N population, n sample)
μ ("mu") / $\bar{x}$ ("x-bar")	the mean (average): μ population, $\bar{x}$ sample
σ ("sigma") / s	standard deviation = typical distance from the mean; σ^2, s^2 = variance (SD²)
z	z-score = how many SDs a value sits above (+) or below (−) the mean
P(E), E^c, N(E)	probability of event E (0 = impossible … 1 = certain); E^c = "not E"; N(E) = # of outcomes in E, N(S) = # in the sample space
P(F | E)	probability of F **given** you already know E happened
X / x, P(x)	X = the random variable (the number an experiment produces); x = one value; P(x) = P(X = x); μ_X, σ_X = its mean and SD; E(X) = expected value = μ_X
n, p, 1 − p; ${}_nC_r$, ${}_nP_r$, n!	binomial: # of trials, P(success), P(failure); ways to pick r of n (order doesn't / does matter); n! = n(n−1)⋯1

Greek letter or capital N → population. Plain letter or small n → sample. Round probabilities to 4 decimals.

WHICH TOOL? — read the question first

Question says…	Use
table of counts + "or / and / given"	Ch 4 rules on the two-way table ★Q1
"how many ways"	counting: multiply / ${}_nP_r$ / ${}_nC_r$
table of x and P(x), or counts of outcomes	Ch 5: valid? μ = Σx·P(x), σ ★Q2 ★Q3
n trials, probability p, "how many successes"	binomial; μ = np, σ = √(np(1−p)) ★Q4
"uniformly distributed between a and b"	rectangle: fraction of the width ★Q5
"normally distributed, mean μ, SD σ"	z → table → left / right / between ★Q6 ★Q7
"what value is the __th percentile / top 10%"	backwards: area → z → x = μ + zσ ★Q7(d)

EXAM 1 SKILLS THAT COME BACK

$$\mu = \frac{\sum x}{N} \qquad \bar{x} = \frac{\sum x}{n} \qquad \sigma = \sqrt{\frac{\sum (x-\mu)^2}{N}} \qquad s = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}}$$

"sample" → divide by n − 1 · calculator: σx = population SD, Sx = sample SD

- **Mean** = the balance point; **median** = middle of the sorted list (resistant to outliers); **mode** = most frequent value. **SD** = typical distance from the mean: subtract the mean, square (the gaps add to 0 otherwise), average the squares (= variance), √.
- **z-score** $z = \frac{x-\mu}{\sigma}$ = how many SDs above (+) or below (−) the mean; compare different scales: the bigger z did relatively better (if lower is better, the smaller z wins); z beyond ±2 → unusual.
- **Empirical Rule** 68–95–99.7 (picture in the normal section).
- **Quartiles** Q_2 = median, Q_1/Q_3 = medians of the halves; **IQR = Q_3 − Q_1; fences Q_1 − 1.5·IQR and Q_3 + 1.5·IQR; outside → outlier**

CH 4 — PROBABILITY

- **Probability** = the long-run fraction of times something happens (Law of Large Numbers: more trials → the fraction settles at the probability). **0 ≤ P(E) ≤ 1; all outcomes together add to 1. Unusual event** = probability below 0.05.
- **Sample space S** = everything that could happen; an **event** = any part of it. Two coins: S = {HH, HT, TH, TT}.
- **Classical** (equally likely outcomes — dice, cards, coins): count. $P(E) = \frac{N(E)}{N(S)}$ P(two heads) = 1/4; P(at least one tail) = 3/4.
- **Empirical** (from data): $P(E) \approx \frac{\text{frequency of } E}{\text{number of trials}}$ 22 of 200 carpool → 0.11 (interpret: "if many people were selected, about 11 of every 100 would carpool"); 5 walk → 0.025 < 0.05 → unusual. **Subjective** = educated guess.
- **Disjoint** (mutually exclusive) = can't both happen (king AND queen), so P(E and F) = 0. **Independent** = E happening doesn't change P(F) (two coin flips). **Disjoint is NOT independent:** if they are disjoint, knowing E happened means F didn't.

The rules — spot the keyword

NOT: $P(E^c) = 1 - P(E)$ · **AT LEAST ONE:** $P(\text{at least 1}) = 1 - P(\text{none})$
OR: $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$ **(disjoint → the overlap is 0, just add)**
AND, independent: $P(E \text{ and } F) = P(E) \cdot P(F)$ · **AND, general:** $P(E \text{ and } F) = P(E) \cdot P(F \mid E)$
GIVEN: $P(F \mid E) = \frac{P(E \text{ and } F)}{P(E)} = \frac{N(E \text{ and } F)}{N(E)}$ ·
Independent? check $P(F \mid E) = P(F)$: equal → independent, different → dependent



Cards: 52 = 4 suits × 13 ranks; 4 of each rank, 13 per suit, 12 face cards (J, Q, K). P(king) = 4/52; P(K or Q or J) = 12/52 (disjoint → add); P(king or ♦) = 4/52 + 13/52 − 1/52 = 16/52 = **0.3077**. **Two picks, no replacement:** the 2nd pick drops top and bottom by 1 → P(2 kings) = 4/52 · 3/51 = **0.0045** (with replacement, or a sample under 5% of the population → 4/52 · 4/52).

- **"At least one":** compute the one case you don't want (none) and subtract from 1. P(survive) = 0.9986: P(both of 2 survive) = 0.9986^2 = 0.9972; P(at least one of 1000 dies) = 1 − 0.9986^{1000} = 0.7536. Without replacement from a big group: sample under 5% of the population → treat as independent.

Counting — "how many ways"

- A **sequence of choices**: multiply the options at each step. 3-letter codes: 26·26·26 = 17,576; chair, vice-chair, secretary from 14: 14·13·12 = 2,184.
- **Permutation** = pick r of n, ORDER MATTERS (1st/2nd/3rd). ${}_nP_r = \frac{n!}{(n-r)!}$ ${}_7P_5$ = 7·6·5·4·3 = 2,520 (r factors counting down); ${}_{10}P_3$ = 720; 7 in order: 7! = 5,040; 0! = 1 so ${}_5P_5$ = 5! = 120 and ${}_nC_0 = {}_nC_n$ = 1.
- **Combination** = pick r of n, order DOESN'T matter (committee, hand, sample). ${}_nC_r = \frac{n!}{r!\,(n-r)!}$ ${}_4C_2$ = 24/(2·2) = 6; ${}_{20}C_4$ = 4,845. Probability by counting: P(4 cards all the same rank) = 13 ÷ ${}_{52}C_4$ = 13/270,725 ≈ 0.00005.

CH 5 — DISCRETE RANDOM VARIABLES ★Q2 ★Q3

- **Random variable X** = the number that comes out of a chance experiment. **Discrete** = you count it; its possible values are a list (x = 0, 1, 2, …). **Continuous** = you measure it; its possible values are an interval (time t ≥ 0, mass 0 < m ≤ 16 oz). **Probability distribution** = every value X can take with its probability.

VALID only if (1) every $0 \le P(x) \le 1$ AND (2) $\sum P(x) = 1$

$$\mu_X = E(X) = \sum [x \cdot P(x)]$$
$$\sigma_X = \sqrt{\sum [(x-\mu)^2 \cdot P(x)]}$$

Mean of a distribution = weighted average (each value counts as much as its probability) = the long-run average per trial. **SD** = typical distance from the mean, each squared gap weighted by its probability.

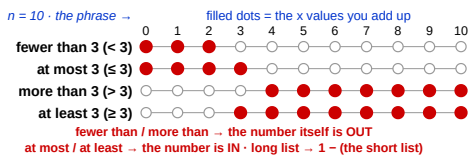
Q: "Insurance: premium $100, pays $10,000 if the client dies, P(dies) = 0.0014. Expected profit?" **A:** build X = profit from the words: lives → +$100 with 1 − 0.0014 = 0.9986; dies → 100 − 10,000 = −$9,900 with 0.0014 (sum = 1 ✓). E(X) = 100(0.9986) − 9900(0.0014) = 99.86 − 13.86 = **$86** per policy (positive → keep selling).

CH 5 (cont.) — BINOMIAL ★Q4

Binomial experiment = the same yes/no trial repeated: (1) FIXED number of trials n, (2) independent, (3) two outcomes (success / failure), (4) SAME p each time. X = number of successes. Not binomial if n isn't fixed ("until…") or picks are without replacement from a small group.

$$P(x) = {}_nC_x\, p^x (1-p)^{n-x} \qquad {}_nC_x = \frac{n!}{x!\,(n-x)!}$$
$$\mu = np \qquad \sigma = \sqrt{np(1-p)}$$

Unusual count (when np(1 − p) ≥ 10): below μ − 2σ or above μ + 2σ · Excel: =BINOM.DIST(x, n, p, FALSE); TRUE = P(X ≤ x)



Q: "Binomial or not? (1) Roll a pair of dice 10 times, count sums of 7. (2) Select flights until 10 late ones. (3) Pick 4 of 30 students, count females." **A:** (1) YES: n = 10, p = 6/36, x = 0, 1, …, 10 (x always runs 0 … n). (2) NO, n not fixed. (3) NO, without replacement from 30 (13% > 5%).

CH 6 — CONTINUOUS & NORMAL

- A continuous variable has infinitely many values, so probability = **AREA under the curve** (the pdf: never below 0, total area 1). **P(X = one exact value) = 0** so P(a < X < b) = P(a ≤ X ≤ b).

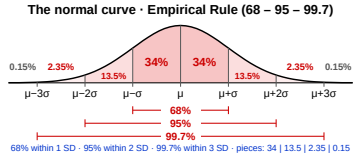
Uniform ★Q5

Uniform on [a, b] = a flat rectangle: every stretch of the same length is equally likely. Probability of an interval = its share of the total width.

$$\text{height} = \frac{1}{b-a} \qquad P(c \le X \le d) = \frac{d-c}{b-a}$$
$$\text{value with area } P \text{ to its left: } x = a + P(b-a)$$

Normal ★Q6 ★Q7

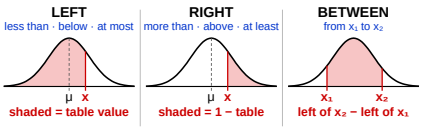
Normal = the bell curve: symmetric around μ, most values near the middle. μ slides it left/right, σ makes it wider or narrower. **Properties:** symmetric about μ; mean = median = mode = μ (the peak); inflection points at μ ± σ; total area 1; tails never touch the axis; Empirical Rule; half the area on each side of μ.



Standard normal Z = μ = 0, σ = 1. **The trick:** convert any value to a z-score and use ONE table (Table V), which gives the area to the LEFT of z.

$$z = \frac{x-\mu}{\sigma} \text{ (round to 2 decimals)} \qquad x = \mu + z\sigma$$

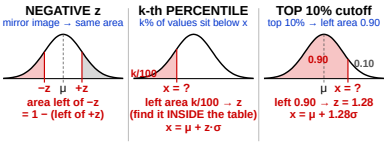
LEFT ("less than / below"): table value · RIGHT ("more than / above"): 1 − table value
BETWEEN x_1 and x_2: (left of z_2) − (left of z_1) · negative z: left of −z = 1 − (left of +z)



Steps: draw the bell, mark μ and x, shade → z = (x − μ)/σ to 2 decimals → table: row = first two digits (1.6), column = last digit (.07) = area to the LEFT → left: done · right: 1 − area · between: subtract the two lefts.

Backwards — find x from a percent / percentile ★Q7(d)

The area is GIVEN; run the steps backwards: words → area to the LEFT ("top 10%" → 0.90, "bottom 5%" → 0.05, "k-th percentile" → k/100, "middle 80%" → 0.10 and 0.90) → find it INSIDE the table (closest entry; exactly halfway → average the two z's: 0.95 → 1.645, 0.995 → 2.575) → read z from the edges → x = μ + zσ.



Percentile → z: 1st −2.33 · 2nd −2.05 · 2.5th −1.96 · 5th −1.645 · 10th −1.28 · 20th −0.84 · 25th −0.67 · 50th 0 · 75th 0.67 · 80th 0.84 · 90th 1.28 · 95th 1.645 · 97.5th 1.96 · 98th 2.05 · 99th 2.33
Middle 80% ↔ z = ±1.28 · middle 90% ↔ ±1.645 · middle 95% ↔ ±1.96 · middle 99% ↔ ±2.575

- **"Give two interpretations" of an area:** "the area to the right of 200 is 0.2903" means (1) the **proportion** of the population above 200 is 0.2903 (29.03%) AND (2) the **probability** that one randomly selected person is above 200 is 0.2903.
- **Is it normal?** histogram bell-shaped, one peak, symmetric, few outliers → yes; a **normal probability plot** (observed values vs the z-scores expected if normal) close to a straight line → yes; a long tail or a bent plot → not normal.

DON'T LOSE POINTS

- "Or" → subtract the overlap once. "And" → the one cell. "Given" → divide by the given group, not the grand total. Σ P(x) = 1 and each P(x) 0–1 before calling a distribution valid.
- "Fewer than 3" stops at 2; "at least 3" = 1 − P(X ≤ 2); "at most" includes the number. Binomial σ needs the √. Uniform: length ÷ total width, never z.
- Normal: z to 2 decimals; table = area to the LEFT; "above" → 1 − left; "between" → big − small. Percentile → area → z (inside the table) → x = μ + zσ. 4 decimals; formula every time.

Standard normal table — area to the LEFT of z (entries 0.xxxx; negative z → 1 − value)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	5000	5040	5080	5120	5160	5199	5239	5279	5319	5359
0.1	5398	5438	5478	5517	5557	5596	5636	5675	5714	5753
0.2	5793	5832	5871	5910	5948	5987	6026	6064	6103	6141
0.3	6179	6217	6255	6293	6331	6368	6406	6443	6480	6517
0.4	6554	6591	6628	6664	6700	6736	6772	6808	6844	6879
0.5	6915	6950	6985	7019	7054	7088	7123	7157	7190	7224
0.6	7257	7291	7324	7357	7389	7422	7454	7486	7517	7549
0.7	7580	7611	7642	7673	7704	7734	7764	7794	7823	7852
0.8	7881	7910	7939	7967	7995	8023	8051	8078	8106	8133
0.9	8159	8186	8212	8238	8264	8289	8315	8340	8365	8389
1.0	8413	8438	8461	8485	8508	8531	8554	8577	8599	8621
1.1	8643	8665	8686	8708	8729	8749	8770	8790	8810	8830
1.2	8849	8869	8888	8907	8925	8944	8962	8980	8997	9015
1.3	9032	9049	9066	9082	9099	9115	9131	9147	9162	9177
1.4	9192	9207	9222	9236	9251	9265	9279	9292	9306	9319
1.5	9332	9345	9357	9370	9382	9394	9406	9418	9429	9441
1.6	9452	9463	9474	9484	9495	9505	9515	9525	9535	9545
1.7	9554	9564	9573	9582	9591	9599	9608	9616	9625	9633
1.8	9641	9649	9656	9664	9671	9678	9686	9693	9699	9706
1.9	9713	9719	9726	9732	9738	9744	9750	9756	9761	9767
2.0	9772	9778	9783	9788	9793	9798	9803	9808	9812	9817
2.1	9821	9826	9830	9834	9838	9842	9846	9850	9854	9857
2.2	9861	9864	9868	9871	9875	9878	9881	9884	9887	9890
2.3	9893	9896	9898	9901	9904	9906	9909	9911	9913	9916
2.4	9918	9920	9922	9925	9927	9929	9931	9932	9934	9936
2.5	9938	9940	9941	9943	9945	9946	9948	9949	9951	9952
2.6	9953	9955	9956	9957	9959	9960	9961	9962	9963	9964
2.7	9965	9966	9967	9968	9969	9970	9971	9972	9973	9974
2.8	9974	9975	9976	9977	9977	9978	9979	9979	9980	9981
2.9	9981	9982	9982	9983	9984	9984	9985	9985	9986	9986
3.0	9987	9987	9987	9988	9988	9989	9989	9989	9990	9990

Reading it: z = 1.67 → row **1.6**, column **.07** → 0.9525. z = −1.67 → 1 − 0.9525 = 0.0475. **Backwards:** area 0.98 → closest entry 0.9798 → row 2.0, column .05 → z = 2.05. z = 3.1 → .9990, 3.2 → .9993, 3.3 → .9995, 3.4 → .9997, above → ≈ 1.